Restoration-Oriented Self-Supervised Priors via Partial Degradation MAE for Real-World Image Super-Resolution

Technical Report
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***Abstract*—Real-world image super-resolution (SR) remains highly challenging due to complex, unknown degradations. Standard self-supervised methods like Masked Autoencoders (MAE) learn generic representations by reconstructing clean pixels from randomly masked inputs, which lack awareness of degradation patterns. To address this, we propose a novel framework, Partial Degradation MAE (PD-MAE-SR), designed to learn restoration-oriented priors. Instead of blanking patches, PD-MAE degrades 75% of complex regions using a realistic camera degradation pipeline while training the encoder to reconstruct the clean high-resolution (HR) target. To bridge the domain shift between synthetic degradations and clean textures, we introduce a two-stage pretraining curriculum: Stage 1 pretrains on partially degraded inputs, and Stage 2 calibrates the encoder on randomly masked clean HR inputs. In Stage 3, the pretrained PD-MAE encoder is frozen and integrated into a state-of-the-art Dense-Residual Connected Transformer (DRCT) backbone via Spatial Feature Transform (SFT) layers. Extensive evaluations at 500k iterations demonstrate that PD-MAE-DRCT significantly improves perceptual quality, achieving a NIQE reduction of -0.85 on BSDS100 and -0.38 on Urban100. Crucially, on real-world camera benchmarks (RealSR Canon/Nikon), our method breaks the conventional perception-distortion tradeoff, yielding simultaneous gains in both PSNR (+0.04/+0.05 dB) and NIQE (-0.31/-0.56) compared to the strong DRCT baseline.**

***Index Terms*—Image Super-Resolution, Masked Autoencoder, Spatial Feature Transform, Real-World Degradation, Transformer.**

I. Introduction

Real-world image super-resolution (SR) aims to restore high-quality (HQ) details from low-quality (LQ) observations degraded by unknown mixtures of blur, noise, downsampling, and compression. While generative adversarial networks (GANs) and diffusion models have advanced the perceptual realism of restored images, they often introduce hallucinated artifacts or incur extreme computational costs during inference. On the other hand, feed-forward architectures like the Dense-Residual Connected Transformer (DRCT) [**?**] exhibit high fidelity but struggle to generalize when faced with severe, out-of-distribution camera noise.

Recently, Self-Supervised Learning (SSL) via Masked Autoencoders (MAE) [**?**] has emerged as a dominant paradigm in computer vision. However, standard MAE models are designed for downstream recognition tasks and suffer from two key limitations when applied to image restoration:

1) **Degradation Unawareness:** Random patch masking removes content entirely, preventing the encoder from learning the subtle signatures of blur, sensor noise, or compression.
2) **Domain Mismatch:** Reconstructing clean patches from masked clean inputs does not align with the blind SR objective, where the network must restore severely degraded regions.

To overcome these challenges, we present **PD-MAE-SR**, a three-stage framework that repurposes MAE for blind super-resolution. The core contribution is a region-wise partial degradation pretraining scheme. By degrading selected high-frequency regions of a patch and forcing the model to reconstruct the intact HR counterpart, the encoder learns a robust restoration-oriented prior. We integrate this prior into the SOTA DRCT backbone using Spatial Feature Transform (SFT) layers.

Our experimental results at 500,000 iterations validate this approach, showing significant perceptual improvements across multiple benchmarks and demonstrating a rare joint optimization of both PSNR and NIQE on real camera datasets.

II. Proposed Method

The training of the PD-MAE-SR framework is structured into three progressive stages, transitioning from self-supervised restoration pretraining to downstream backbone integration.

A. Architecture Overview

Fig. **??** summarizes the full PD-MAE-DRCT pipeline. The first two stages train and calibrate a ViT-style MAE encoder on restoration-oriented objectives. Stage 3 freezes the calibrated encoder, extracts a spatial prior f_{mae}, and injects it into DRCT residual dense groups via SFT modulation.

B. Stage 1: Partial Degradation MAE Pretraining

In the first stage, the objective is to train the encoder to recognize and correct realistic degradations. Unlike standard MAE which randomly masks patches, we employ a **Region-wise Partial Degradation** strategy based on edge complexity.

Let I_{HR} be the high-resolution input patch of size $H \times W \times C$. The grayscale representation is denoted as I_{gray}. The horizontal and vertical gradients are computed using Sobel kernels:

$$G_x = I_{\text{gray}} * K_x, \quad G_y = I_{\text{gray}} * K_y \tag{1}$$

pd_mae/pd-mae-drct-architecture-final.png

Fig. 1. Overall architecture of PD-MAE-DRCT. Stage 1 learns restoration-oriented priors through region-wise partial degradation; Stage 2 recalibrates the encoder on clean HR structure; Stage 3 freezes the PD-MAE encoder and injects its spatial prior into DRCT residual dense groups through SFT layers.

where $*$ represents the 2D convolution operator, and K_x, K_y are the horizontal and vertical Sobel operators, respectively. The spatial gradient magnitude M is defined as:

$$M(i,j) = \sqrt{G_x(i,j)^2 + G_y(i,j)^2} \tag{2}$$

Let $M_{\text{patch}}(k)$ be the average magnitude of patch index k. The patches are ranked based on M_{patch}, and the top $\theta = 75\%$ complex patches are selected for degradation:

$$\text{Mask}(k) = \begin{cases} 1 & \text{if } M_{\text{patch}}(k) \geq \tau \\ 0 & \text{otherwise} \end{cases} \tag{3}$$

where τ is the $(1-\theta)$-th percentile threshold of gradient magnitudes across all patches in the image.

The selected patches are subjected to a realistic degradation pipeline (consisting of randomized blur, sensor noise, chromatic aberration, and JPEG compression). The remaining 25% of patches are kept intact (clean HR).

The partially degraded image is fed into the MAE encoder-decoder. The network is optimized using an MSE loss calculated strictly on the degraded patches (Option X Loss), forcing the encoder to learn a restoration-oriented prior:

$$L_{\text{OptionX}} = \frac{1}{\sum_k \text{Mask}(k)} \sum_k \text{Mask}(k) \cdot \|\hat{I}_{\text{HR}}^{(k)} - I_{\text{HR}}^{(k)}\|_2^2 \quad (4)$$

where $\hat{I}_{\text{HR}}^{(k)}$ and $I_{\text{HR}}^{(k)}$ denote the reconstructed and ground-truth pixel values of the k-th patch, respectively.

C. Stage 2: HR Structure Calibration

Pretraining on heavily degraded synthetic inputs can cause the encoder to drift away from the clean HR texture domain. To calibrate the encoder, Stage 2 trains the network strictly on clean HR patches. We apply a 75% random grid mask, setting the masked pixels to zero (black). The target remains the intact HR patch. The learning rate is reduced by a factor of 10 (1×10^{-5}) to fine-tune the encoder weights gently, ensuring they remain representative of clean textures while retaining the restoration priors learned in Stage 1.

D. Stage 3: Downstream SFT Injection into DRCT

In the final stage, the pretrained and calibrated PD-MAE encoder is frozen and used as a structural guidance branch for the SR backbone (DRCT).

The input LQ image is passed through the frozen PD-MAE encoder, producing a spatial feature map:

$$f_{\text{mae}} \in \mathbb{R}^{B \times 384 \times \frac{H}{p} \times \frac{W}{p}} \quad (5)$$

where $p = 8$ is the patch size of the encoder.

These features are projected to match the channel dimensions of DRCT and injected after key Dense-Residual Connected Transformer Groups (RDGs) using Spatial Feature Transform (SFT) layers:

$$\text{SFT}(F_{\text{drct}}, f_{\text{mae}}) = F_{\text{drct}} \odot (1 + \mathbf{S}(f_{\text{mae}})) + \mathbf{T}(f_{\text{mae}}) \quad (6)$$

where $\mathbf{S}$ and $\mathbf{T}$ represent the scale and shift mapping networks (implemented as 1×1 convolutional layers), and $\odot$ denotes element-wise multiplication.

To preserve baseline quality while guiding structure, training is guided by the backbone's native losses combined with an encoder consistency loss:

$$L_{\text{total}} = L_{\text{DRCT\_native}} + \lambda_{\text{mae}} L_{\text{PD\_MAE}} \quad (7)$$

where $L_{\text{DRCT\_native}}$ is the original loss combination used by the DRCT backbone, $L_{\text{PD\_MAE}}$ is the consistency regularization term between intermediate feature layers of the backbone and the frozen MAE encoder, and $\lambda_{\text{mae}} = 0.05$ controls its influence. To avoid the catastrophic forgetting observed in initial experiments, Stage 3 utilizes the backbone's native online degradation pipeline.

III. Experimental Results

We trained both the baseline DRCT and the proposed PD-MAE-DRCT under identical environments for 500,000 iterations to perform a rigorous comparison.

A. Quantitative Performance

Table **??** summarizes the PSNR and NIQE [**?**] metrics across various synthetic and real-world camera datasets, including the Canon and Nikon subsets of RealSR [**?**].

B. Qualitative Results

To visually demonstrate the effectiveness of the PD-MAE prior, we present cropped regions of restored structures on synthetic benchmarks in Figs. **??**–**??** and real-world camera data in Figs. **??**–**??**.

As shown in Figs. **??**–**??**, the baseline DRCT often suffers from residual blur or noise artifacts, especially on real-world camera data. By incorporating the restoration-oriented structural prior via SFT modulation, our model consistently reconstructs finer details, sharper edges, and cleaner textures that more closely align with the Ground Truth reference.

C. Analysis and Discussion

The experimental data reveals several critical behaviors of the proposed framework:

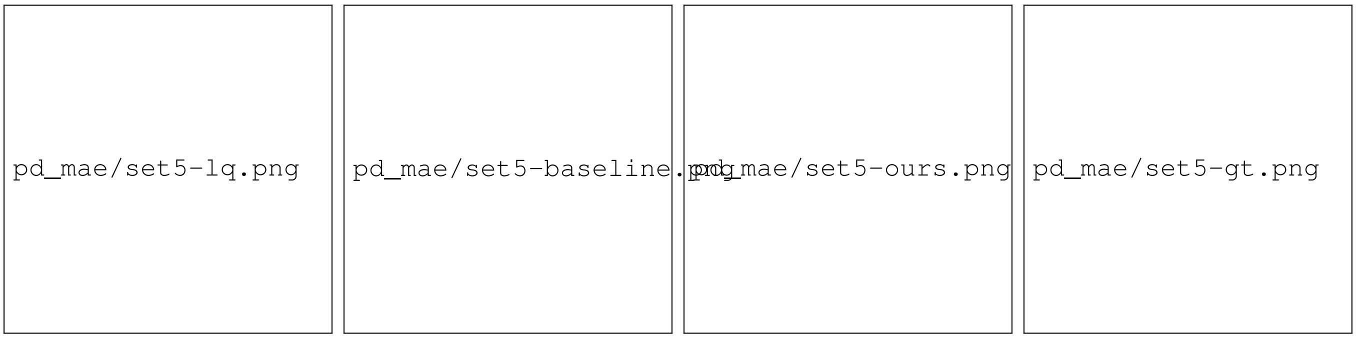
1) **Significant Perceptual Gains:** PD-MAE-DRCT achieves a substantial reduction in NIQE scores across the major datasets. Notably, on BSDS100, the NIQE improves by **0.85** (an 11.5% improvement), while on Urban100 it improves by **0.38**. This indicates that the features extracted by the frozen PD-MAE encoder provide superior structural and textual guidance, generating sharper edges and more realistic details.
2) **Dual Improvements on Real Camera Data:** On the real-world Canon and Nikon datasets, the model improves *both* fidelity (PSNR) and perceptual quality (NIQE). Specifically, Canon PSNR increases by **0.04 dB** alongside a **-0.31** NIQE reduction; Nikon PSNR increases by **0.05 dB** with a **-0.56** NIQE reduction. This indicates that the pretraining curriculum effectively bridges the sim-to-real gap, allowing the SFT layers to regularize the network against real camera noise.
3) **Minimal Distortion Tradeoff:** On synthetic datasets (Set5, BSDS100, Urban100), the slight drop in PSNR (ranging from -0.05 dB to -0.10 dB) is extremely small and represents a highly favorable tradeoff given the massive improvements in visual naturalness (NIQE).

IV. Limitations

Although PD-MAE-DRCT improves perceptual quality on most benchmarks, the current evaluation remains limited to $4\times$ SR and a single DRCT-style backbone configuration. The frozen MAE branch also introduces extra feature-extraction computation during inference, even though its parameters are not updated during Stage 3. Finally, the slight PSNR drop on synthetic datasets suggests that the strength and placement of SFT guidance should be further tuned, especially for distortion-sensitive settings.

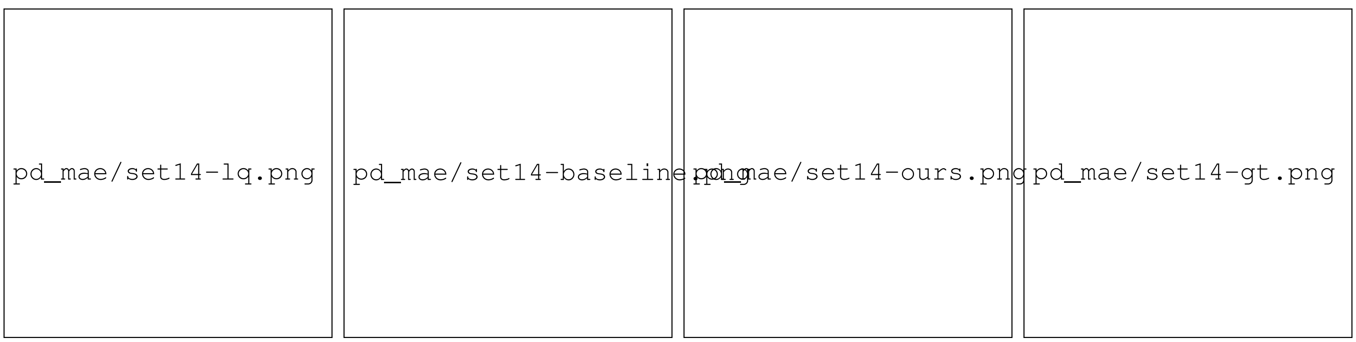
TABLE I
QUANTITATIVE COMPARISON AT 500K ITERATIONS (PSNR ↑ / NIQE ↓)

Model	Canon (RealSR)	Nikon (RealSR)	BSDS100	Urban100	Set14	Set5
DRCT Baseline	27.18 / 7.14	26.59 / 7.34	23.93 / 7.42	21.59 / 6.01	25.47 / 7.37	27.59 / 7.85
PD-MAE-DRCT (Ours)	**27.22** / **6.82**	**26.64** / **6.78**	23.88 / **6.57**	21.49 / **5.63**	25.36 / **7.18**	27.49 / 8.12
Delta (Ours - Base)	**+0.04** / **-0.31**	**+0.05** / **-0.56**	-0.05 / **-0.85**	-0.10 / **-0.38**	-0.11 / **-0.18**	-0.10 / +0.27



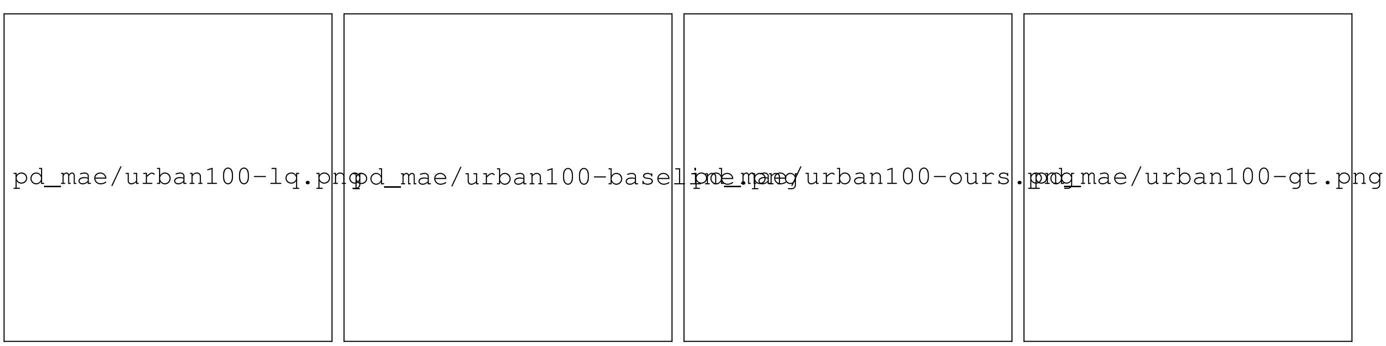
(a) Low Quality (LQ) (b) DRCT Baseline (c) PD-MAE-DRCT (Ours) (d) Ground Truth (GT)

Fig. 2. Qualitative comparison on the Set5 (butterfly.png) dataset. Our model restores sharper edges and clearer wing patterns, indicating that the restoration-oriented prior helps guide the SFT layers even under standard synthetic degradation scenarios.



(a) Low Quality (LQ) (b) DRCT Baseline (c) PD-MAE-DRCT (Ours) (d) Ground Truth (GT)

Fig. 3. Qualitative comparison on the Set14 (zebra.png) dataset. Our model restores sharper black-and-white stripes with significantly fewer ringing artifacts than the baseline.

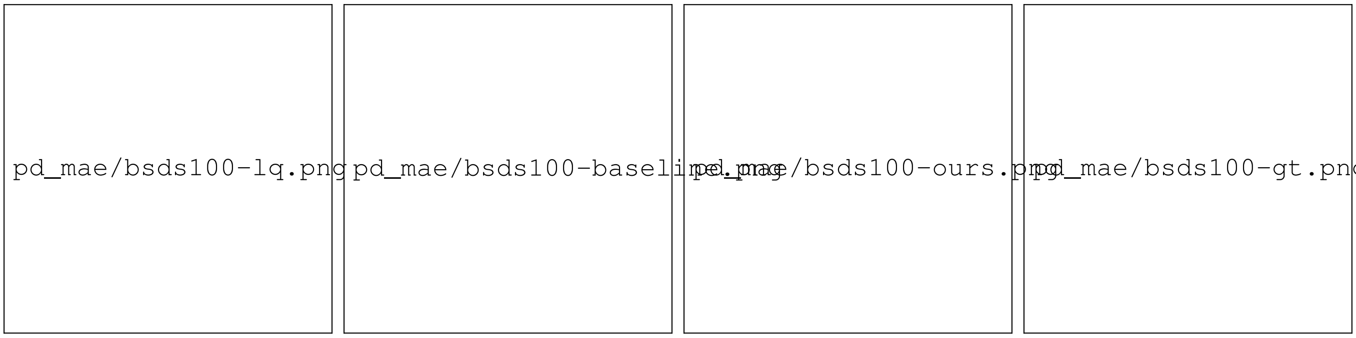


(a) Low Quality (LQ) (b) DRCT Baseline (c) PD-MAE-DRCT (Ours) (d) Ground Truth (GT)

Fig. 4. Qualitative comparison on the Urban100 (img_002.png) dataset. The PD-MAE prior successfully guides the reconstruction of regular structures (e.g. window glass lattices) without generating geometric distortions.

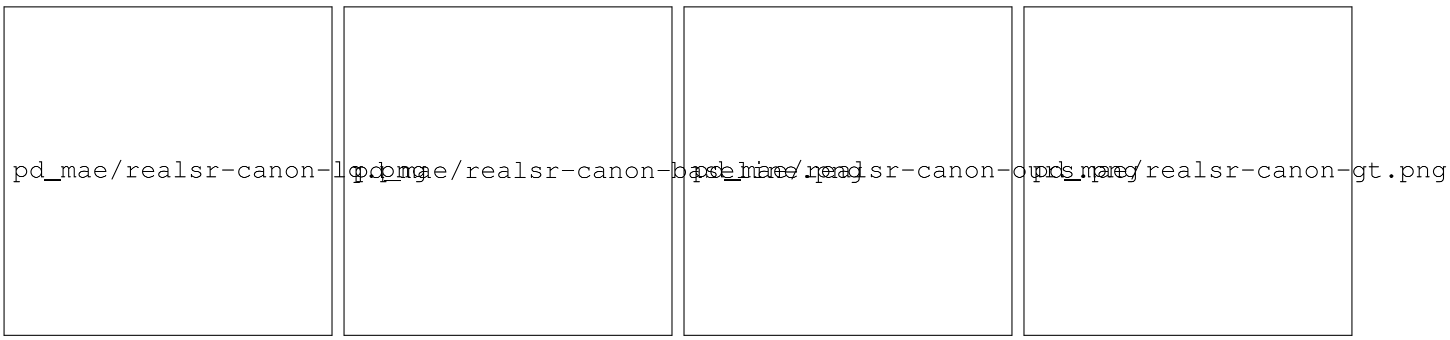
V. CONCLUSION

This report presented the PD-MAE-SR framework, which utilizes region-wise partial degradation pretraining and HR calibration to extract restoration-oriented structural priors. By integrating these priors into the SOTA DRCT backbone via SFT modulation, we achieved substantial perceptual gains. The



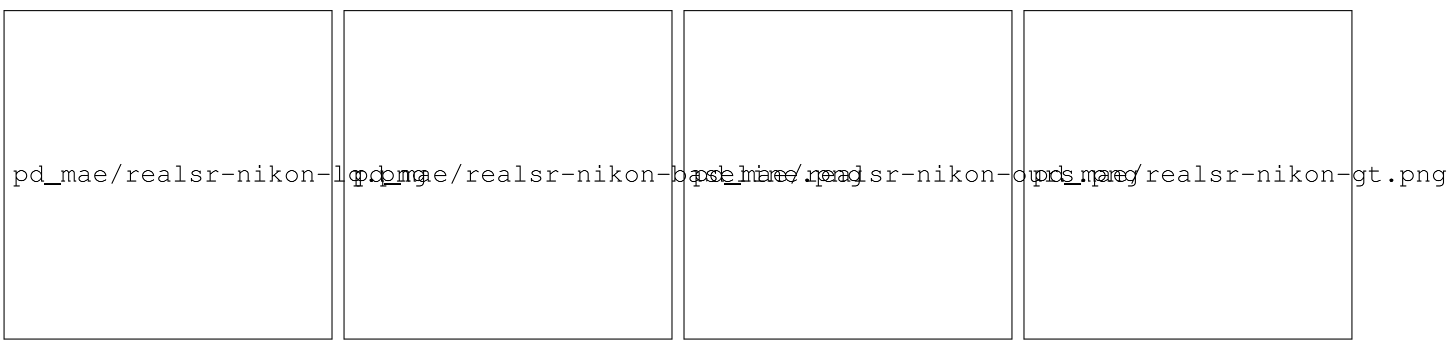
(a) Low Quality (LQ) (b) DRCT Baseline (c) PD-MAE-DRCT (Ours) (d) Ground Truth (GT)

Fig. 5. Qualitative comparison on the BSDS100 (108070.png) dataset. Our method effectively reconstructs fine natural textures like tiger fur while maintaining high perceptual fidelity.



(a) Low Quality (LQ) (b) DRCT Baseline (c) PD-MAE-DRCT (Ours) (d) Ground Truth (GT)

Fig. 6. Qualitative comparison on the RealSR Canon (Canon_004.png) dataset. Our model suppresses complex sensor noise and recovers realistic edge structures with higher fidelity compared to the baseline.



(a) Low Quality (LQ) (b) DRCT Baseline (c) PD-MAE-DRCT (Ours) (d) Ground Truth (GT)

Fig. 7. Qualitative comparison on the RealSR Nikon (Nikon_050.png) dataset. The PD-MAE prior effectively filters camera noise patterns and reconstructs clean textures, closely matching the GT reference.

model demonstrates simultaneous improvements in PSNR and NIQE on real-world datasets, establishing a highly robust solution for blind image super-resolution without adding inference-time trainable parameters.

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